

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.5/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE’25 IIT KANPUR | B2B Brand Equity | Prof.RN Sengupta (May’25-July’25)

Objective	• Identify determinants of B2B brand equity using literature, company data, and test models via ML .
Approach	• Reviewed literature on B2B Brand Equity to map constructs like awareness, loyalty, trust, and reputation. • Applied SEM and multiple logistic regression on 5 firms; analyzed impact of digital marketing using MROI . •Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	• Proposed methods to quantify brand equity using ML models and highlighted digital marketing’s role via MROI .

Key Projects

Financial Markets & Investment Banking | Finlatics 📊 (Dec’24-Feb’25)

- Managed equity portfolio on **BSE 500 Index**; applied portfolio construction diversification, scored **Finscore: 81.16**.
- Conducted sectoral analysis using **Porter’s 5 Forces**; portfolio restructuring and strategic fund positioning; rated **82.12**.
- Completed **Private Equity** and **Investment Banking** project covering investments, fund setup case studies; rated **91.21**.

Hydro Mechanics Of Finvision | Google Developer Group Project| 📊 (Dec’24-Mar’25)

- Built real-time financial data pipelines via **Selenium, Requests, BeautifulSoup** from **Yahoo Finance, NSE**, etc.
- Applied **Black-Scholes, Binomial, Vasicek, CIR, Merton Models**; simulated **GBM, ADF, Weiner** processes for forecasting & risk.
- Used **ML (LSTM, RNN)** for **ETH** prediction; explored quantum models using **TensorFlow Quantum, PennyLane**.

White Paper For Assistive Devices | IITK Consulting Group | Department Of SnT (April’25-June’25)

- Worked under **Dr. Anita Aggarwal (DST, NAMEDD)** on a national white paper for assistive devices (ADs), targeting policy investment.
- Co-authored recommendations using global best practices comparative insights from **USA, Japan, Germany, Denmark**.
- Built global leaderboard, mapped **India’s top 10 medtech states**; analyzed **7+ govt schemes, 20+ VC firms/accelerators/startup studios**.

Technical Skills

- **Programming Languages:** C, C++, Python
- **Software:** Canva, Arduino, MS Excel, Jupyter
- **Libraries:** Numpy, Panda, Tensorflow, Seaborn, Matplotlib, Keras, Scikit-Learn

Relevant Courses

Introduction To Management	Fundamentals of Computing - I	Application of Single Variable Calculus
Ordinary Differential Equations	Linear Algebra	Fundamentals of Computing - II

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team Mar’24-Present

- Contributed to **avionics, structures, propulsion, recovery**; launched rocket to **600 ft** apogee in **July 2024**.
- Managed finances; secured **2 sponsorships**, handled **5 lakh** budget, and oversaw bill clearance.
- Led **30+ juniors**, handled outreach, sponsorship, and participating as Recovery member in **INSPACE Competition**.

Marketing Secretary | Udghosh’24 Aug’24-Oct’24

- Led lead generation and **database management** for **100+ companies** to support sponsorship and marketing outreach.
- Developed strong **communication skills** including cold calling, email writing, and persuasive brand negotiation.

Hospitality Secretary | Inter IIT Sports Meet’24 Nov’24-Dec’24

- Managed hospitality for over **5,000 players** across IITs, ensuring smooth accommodation, meals, and logistics.
- Utilized **Microsoft Excel** for rooming, food tracking, and scheduling to improve efficiency and reduce errors.

Extra-Curricular Activities

- Received **2 LORs,certificates** for outstanding performance in **finlatics financial markets investment banking** project.
- Showcased **Team RaSET** rocketry work at **IITK Samanvay 2024** and **Antarikx’24** through demos and space outreach.
- Completed **NCC ‘C’ Certificate** program, demonstrating leadership, discipline, and endurance.
- Cleared **PDR and CDR** rounds of **INSPACE Model Rocketry Competition** with IITK RaSET team.